

Disciplina: Eletrônica Básica

Aula_4 - Diodos

Ambiente Blackboard

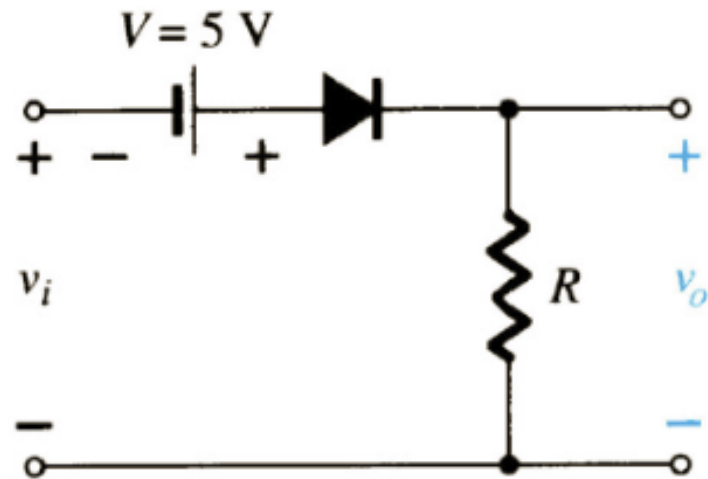
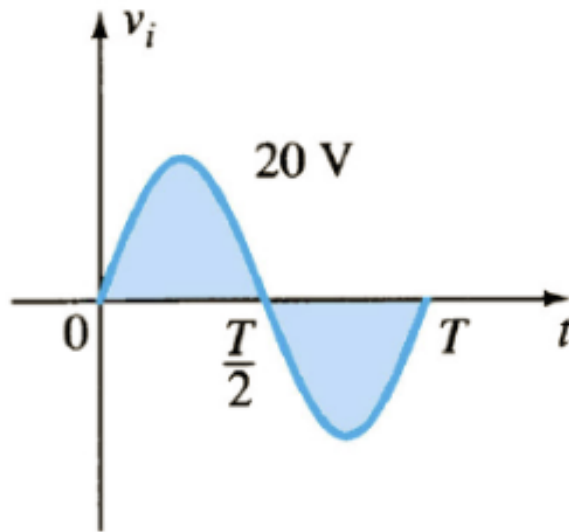
Apresentação

- 1) Ceifadores/Grampeadores - Tarefas
- 2) Retificadores
- 3) Meia onda - Exercícios
- 4) Onda completa CT - Exercícios
- 5) Onda completa Ponte - Exercícios
- 6) Tarefas

Tarefas

Simule os circuitos apresentados nessa aula.

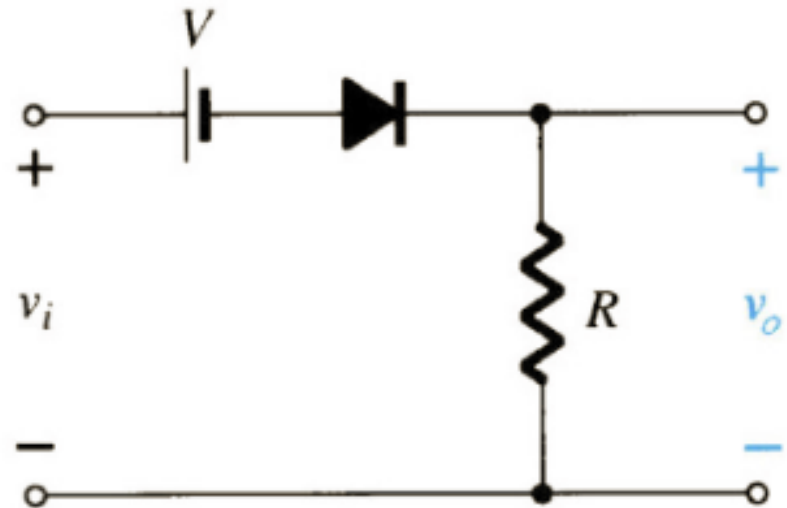
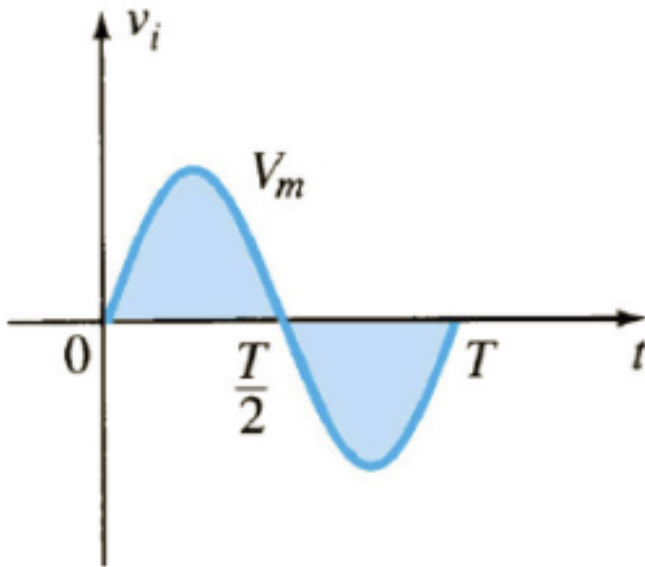
Ceifador série com fonte, exemplo:



Tarefas

Simule os circuitos apresentados nessa aula.

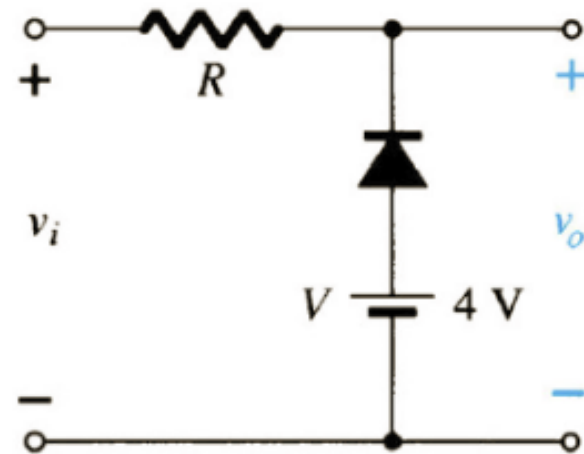
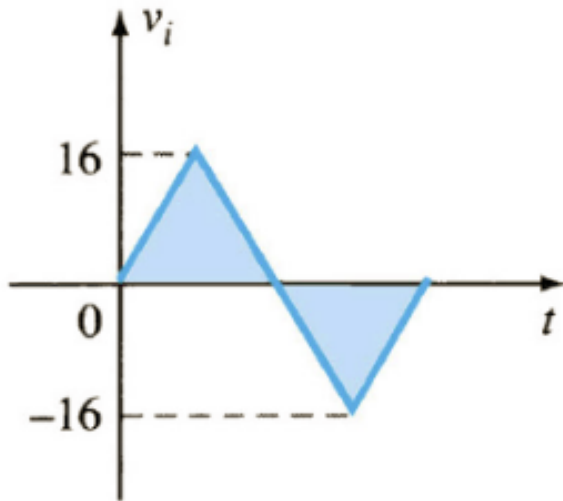
Ceifador série com fonte:



Tarefas

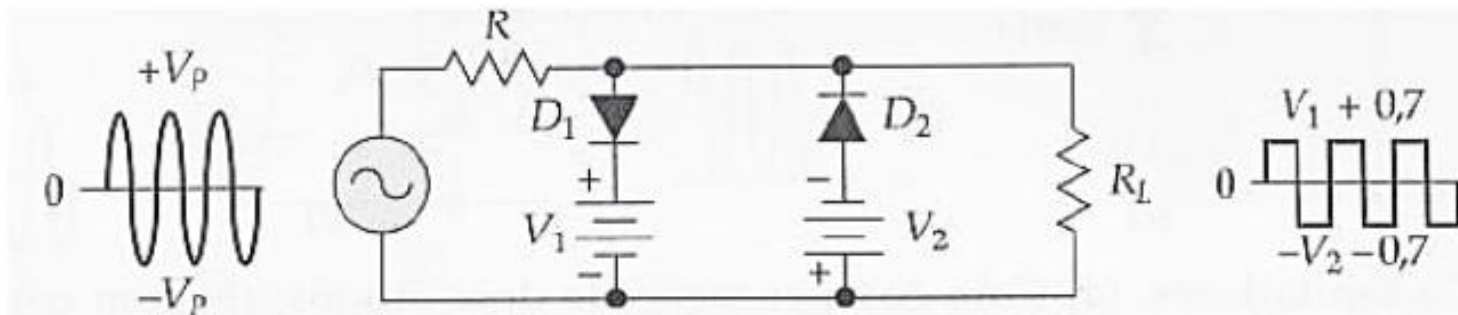
Simule os circuitos apresentados nessa aula.

Ceifador paralelo com fonte, exemplo:



Tarefas

Simule os circuitos apresentados nessa aula.

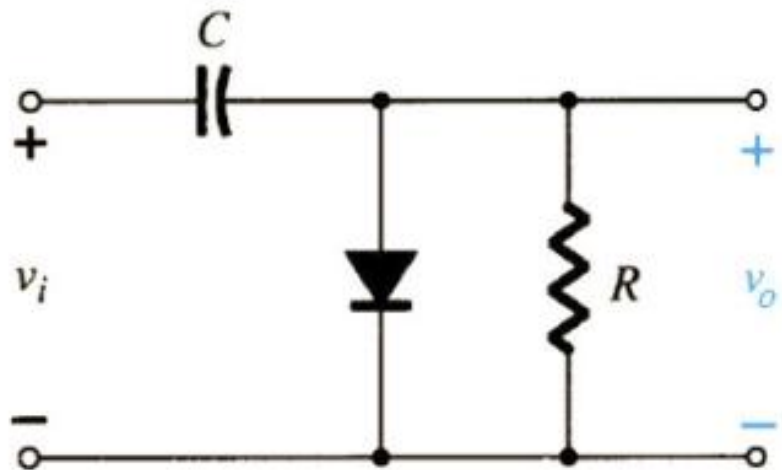
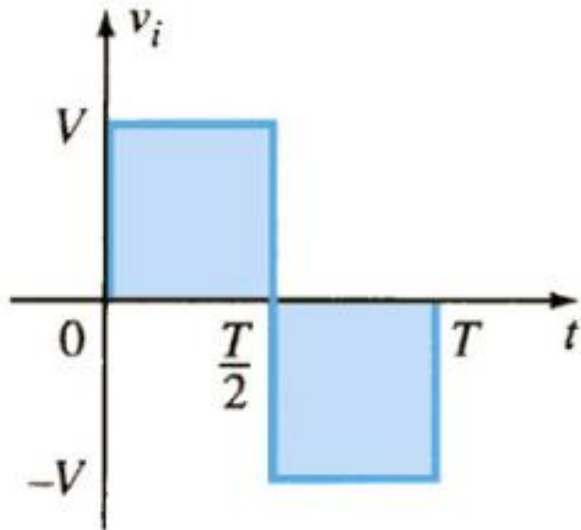


Tarefas

Simule os circuitos apresentados nessa aula.

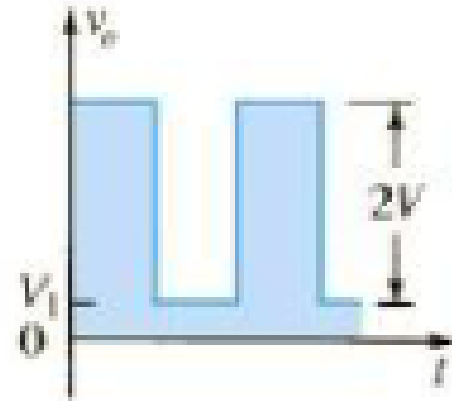
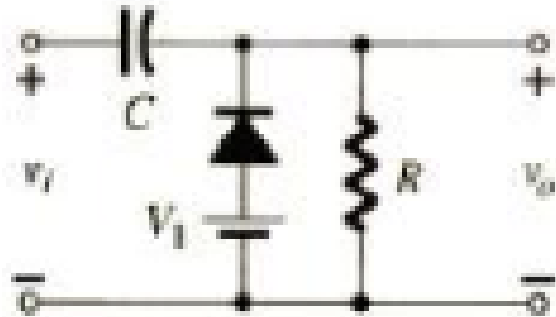
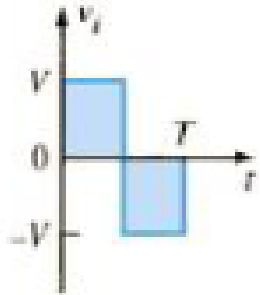
Circuitos grampeadores:

- Tem a capacidade de grampear um sinal em um valor cc diferente.



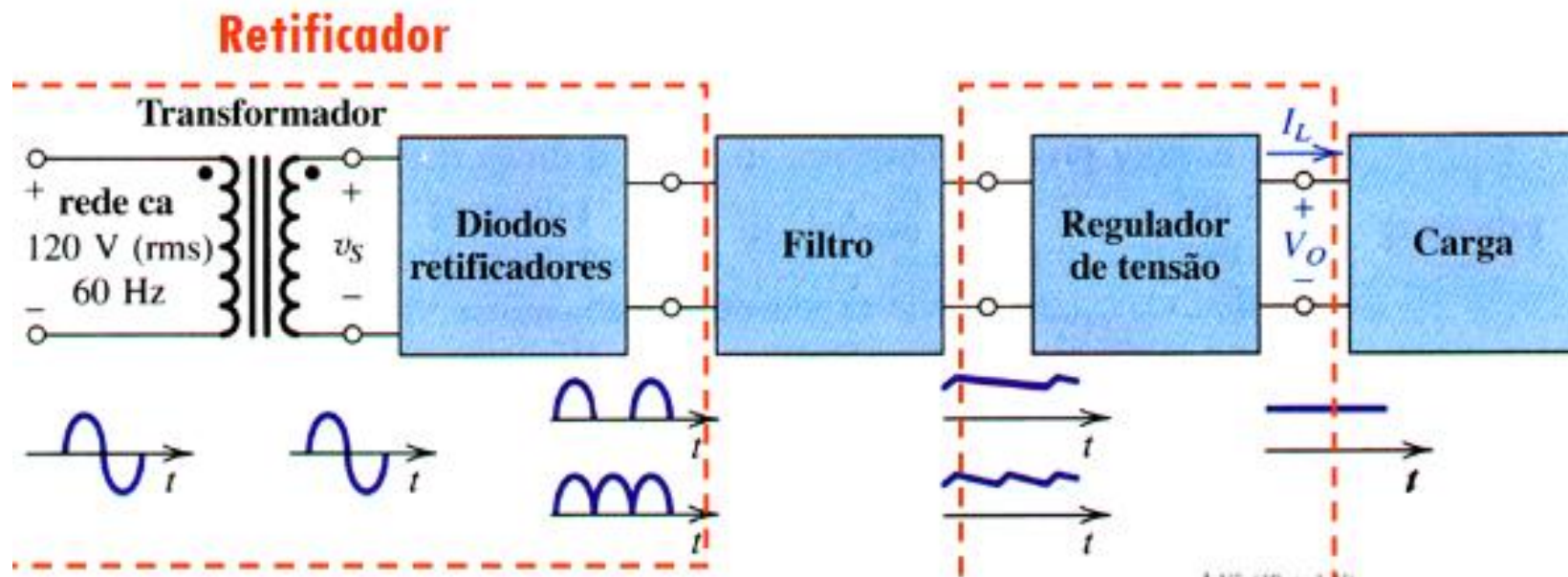
Tarefas

Simule os circuitos apresentados nessa aula.

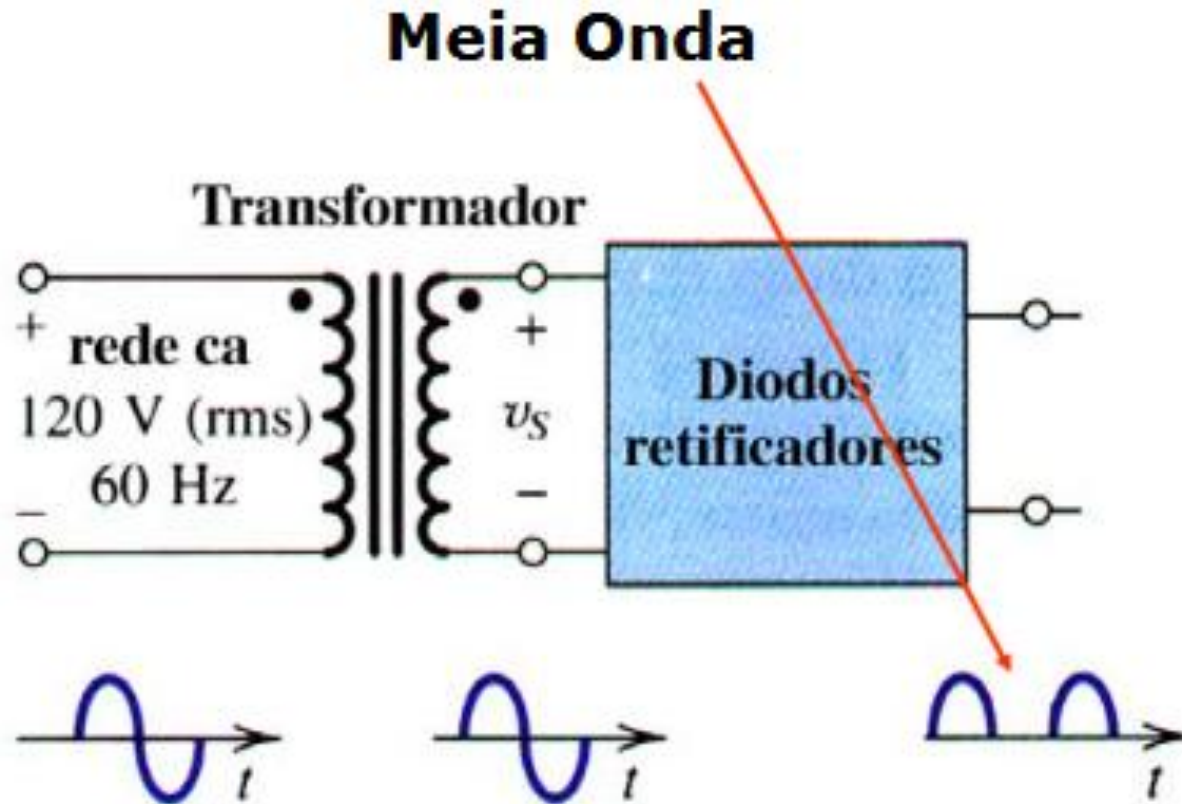


Retificadores

- Circuitos utilizados para conversão de CA/CC

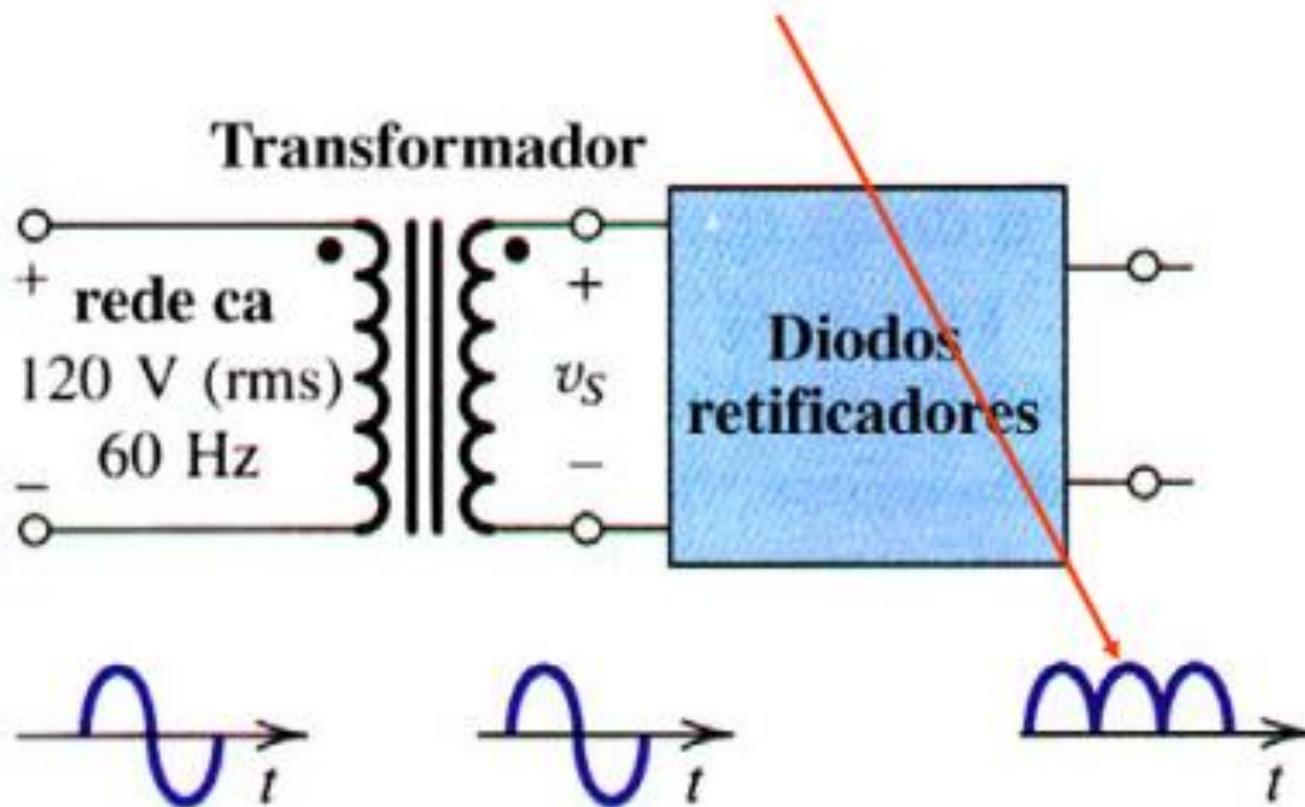


Retificador – Meia Onda



Retificador – Onda Completa

Onda Completa



Retificador – Meia Onda

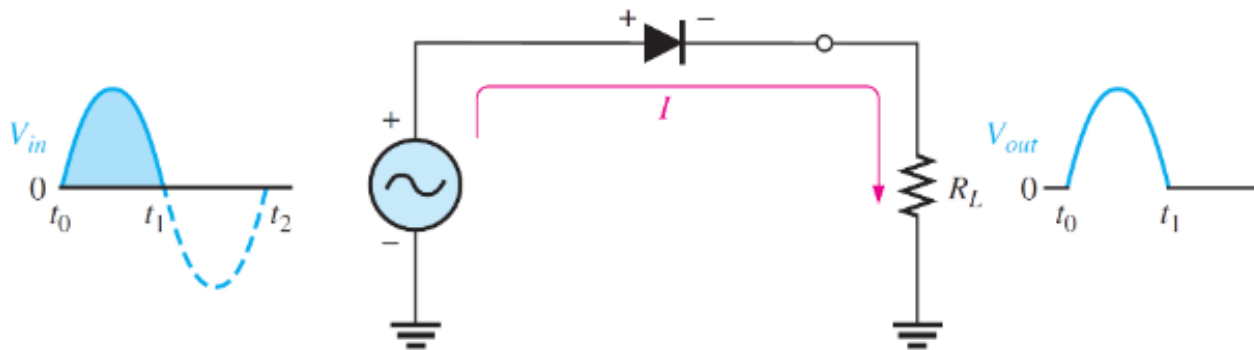
Half-Wave Rectifier Operation

- The process of removing one-half the input signal to establish a dc level is called *half-wave rectification*.

Using the ideal model for the diode:

During the positive half cycle:

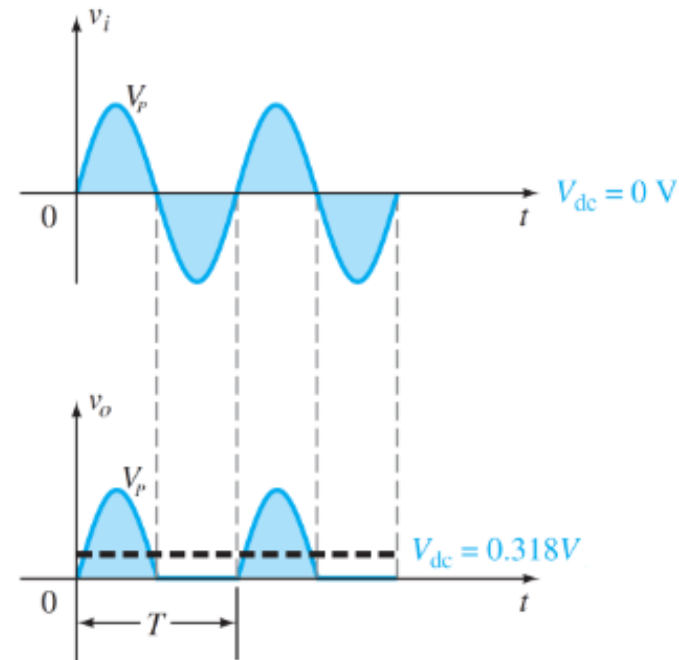
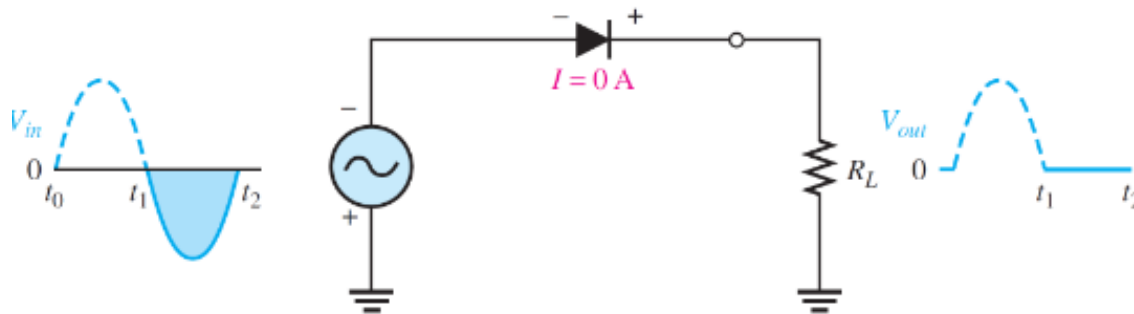
- v_{in} is positive $\rightarrow$ the diode is forward-biased (ON - closed switch).
- The output voltage has the same shape as the positive half-cycle of the input voltage. ($V_{out} = V_{in}$)



Retificador – Meia Onda

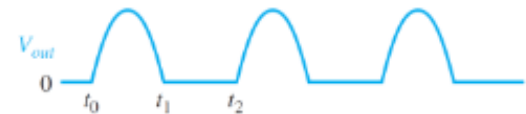
During the negative half cycle:

- v_{in} is negative $\rightarrow$ the diode is reverse-biased.
- the output voltage is 0 V ($V_{out} = 0V$)



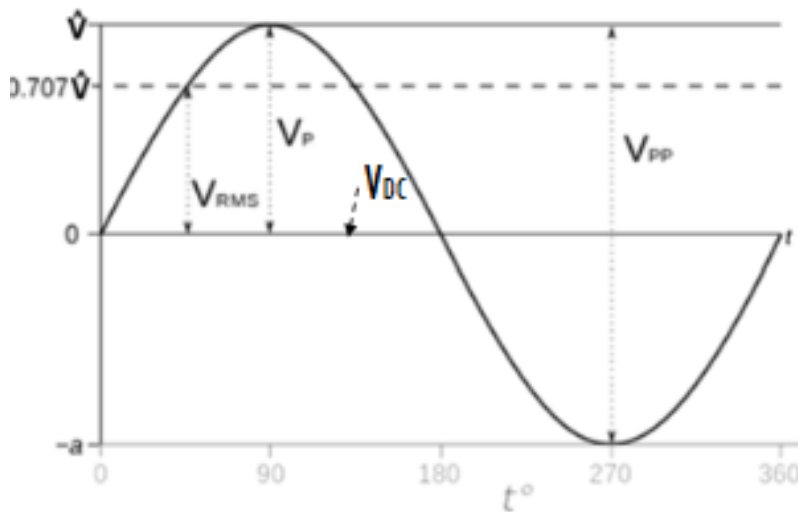
- Since the output does not change polarity it is a pulsating dc voltage with a frequency equal the input frequency

$$\text{HWR: } (f_{out} = f_{in})$$



Revisão – Ondas Senoidais

Ondas Senoidais



$$V_{DC} = \frac{1}{T} \int_0^T V_p \sin(\omega t) dt = 0$$

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^T [V_p \sin(\omega t)]^2 dt} = \frac{V_p}{\sqrt{2}}$$

$V_p = \hat{V} = \text{valor de pico}$

V_{DC} ou $V_m = \text{valor DC ou valor médio}$

$V_{rms} = \text{valor quadrático médio (rms)}$

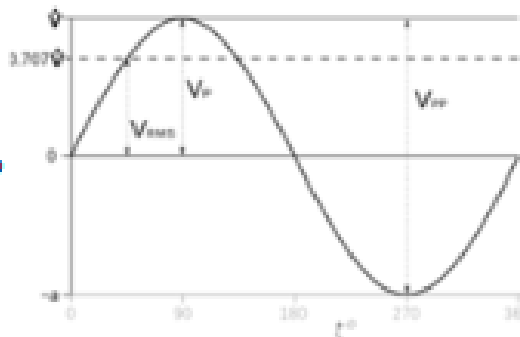
Revisão – Ondas Senoidais

Ondas Senoidais

$V_p = \hat{V}$ = valor de pico

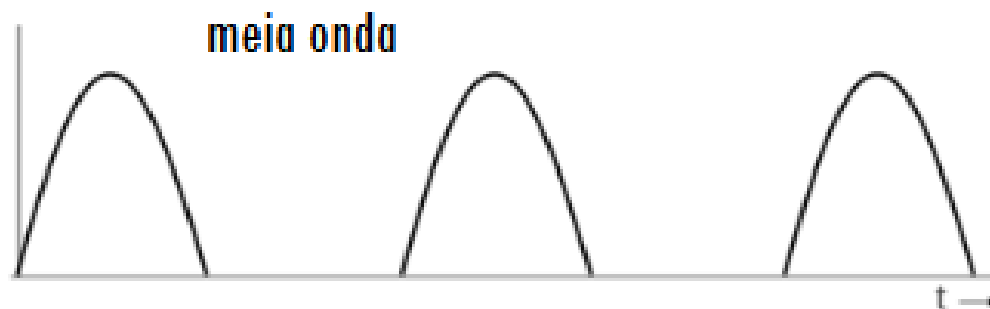
V_{DC} ou V_m = valor DC ou valor médio

V_{rms} = valor quadrático médio (rms)



$$V_{DC} = \frac{1}{T} \int_0^T V_p \sin(\omega t) dt = 0$$

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^T [V_p \sin(\omega t)]^2 dt} = \frac{V_p}{\sqrt{2}}$$



$$V_{DC} = \frac{1}{T} \left[\int_0^{T/2} V_p \sin(\omega t) dt + \int_{T/2}^T 0 dt \right] = \frac{V_p}{\pi}$$

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^{T/2} [V_p \sin(\omega t)]^2 dt} = \frac{V_p}{2}$$

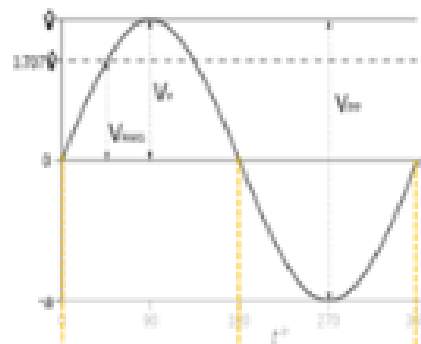
Revisão – Ondas Senoidais

Ondas Senoidais

$V_p = \hat{V}$ = valor de pico

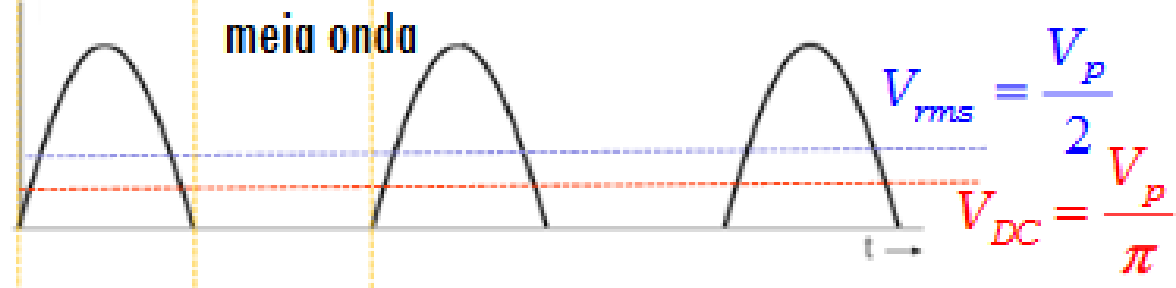
V_{DC} ou V_m = valor DC ou valor médio

V_{rms} = valor quadrático médio (rms)



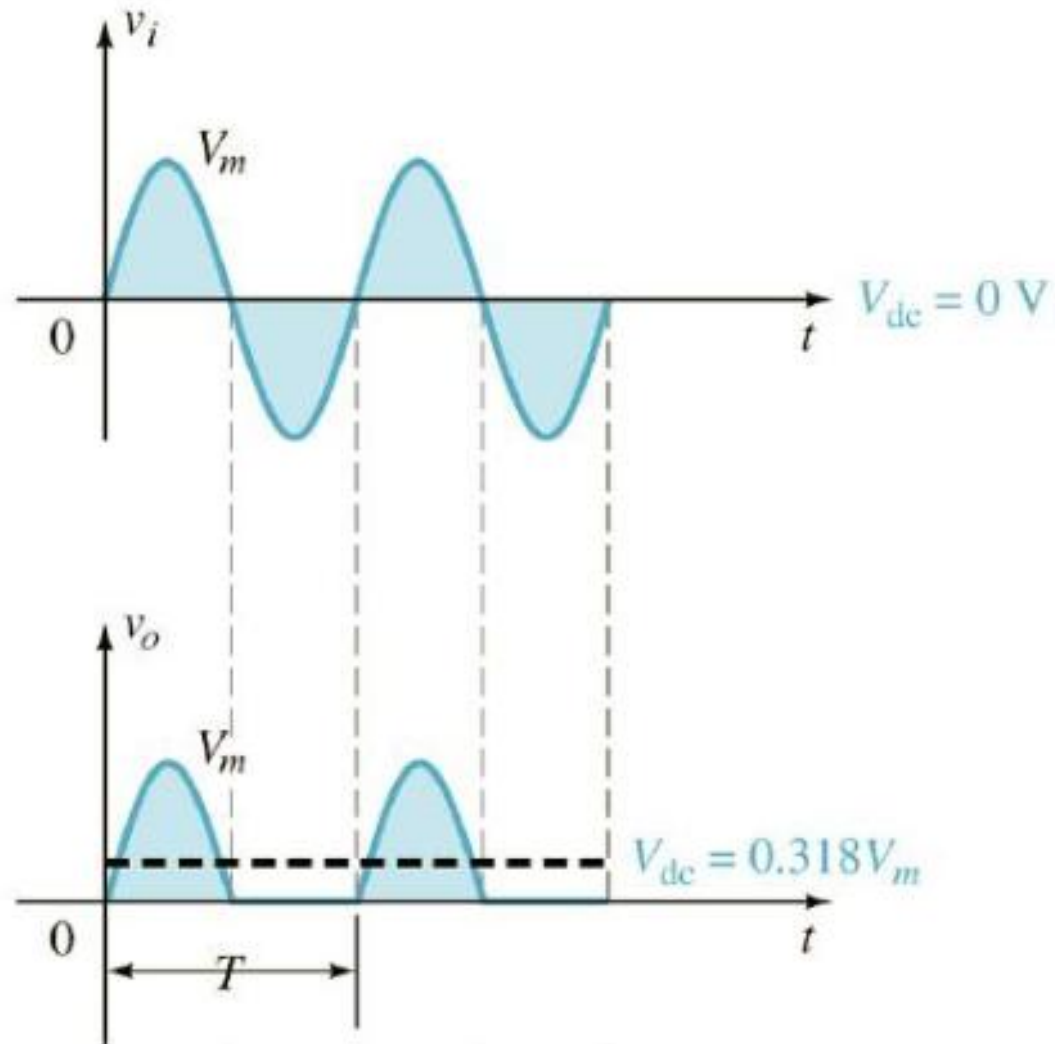
$$V_{DC} = 0$$

$$V_{rms} = \frac{V_p}{\sqrt{2}}$$



Retificador – Meia Onda

Forma de onda resultante:

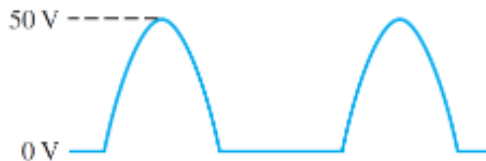


Retificador – Meia Onda

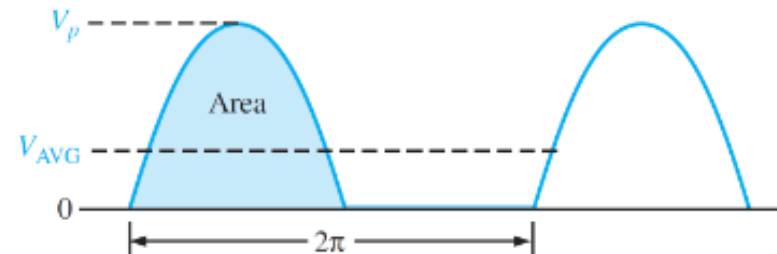
Average Value of the Half-Wave Output Voltage

- The average value of the half-wave rectified output voltage is **the value you would measure on a dc voltmeter**.
- Mathematically, it is determined by finding **the area under the curve over a full cycle**, and then dividing by 2π (the number of radians in a full cycle).

$$V_{AVG} \text{ or } V_{dc} = \frac{V_P}{\pi} = 0.318 V_P$$



$$V_{AVG} = \frac{V_P}{\pi} = \frac{50 \text{ V}}{\pi} = 15.9 \text{ V}$$



Notice that V_{AVG} is 31.8% of V_P .

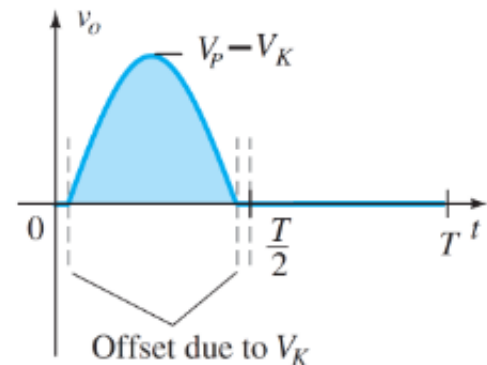
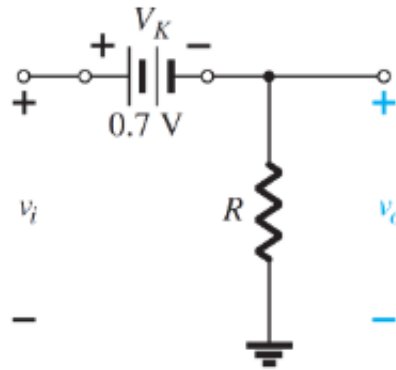
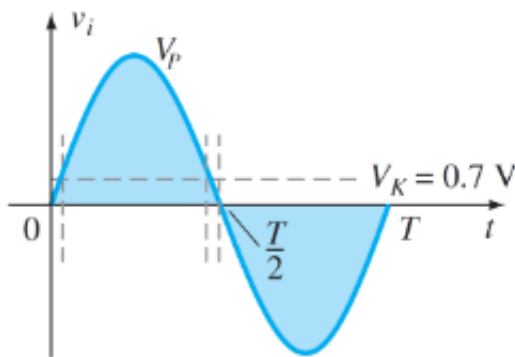
Retificador – Meia Onda

Effect of the Barrier Potential on the Half-Wave Rectifier Output *(Using the practical diode model)*

- When $V_{in} < 0.7$ the diode is OFF (O.C.) and ($V_{out} = 0V$)
- When $V_{in} > 0.7$ the diode is ON and ($V_{out} = V_{in} - 0.7$)
- The peak output ($V_{P(out)} = V_{P(in)} - 0.7$)

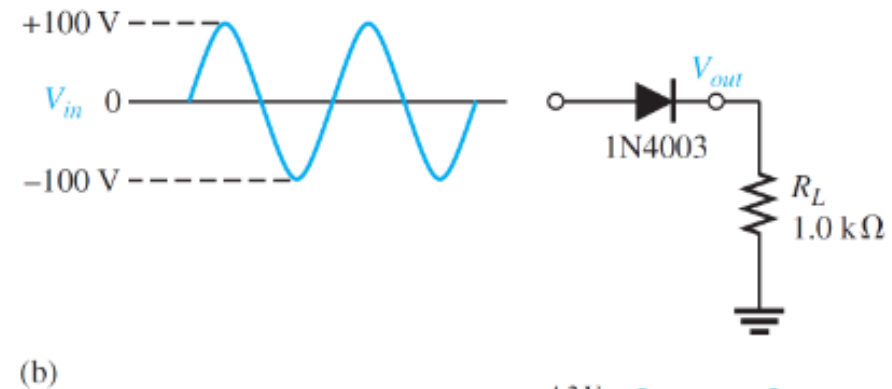
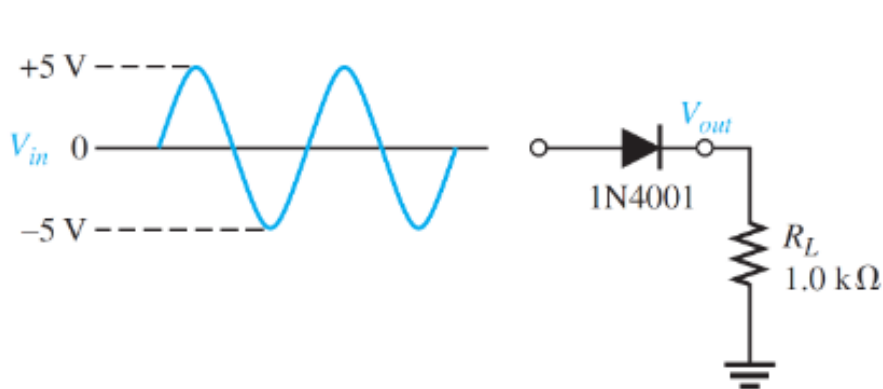
When $V_P \gg 0.7 \rightarrow V_{AVG} \approx \frac{V_P - 0.7}{\pi}$

$V_K = 0.7 V$



Retificador – Meia Onda

EXAMPLE1 Draw the output voltages of each rectifier for the indicated input voltages. The 1N4001 and 1N4003 are specific rectifier diodes.

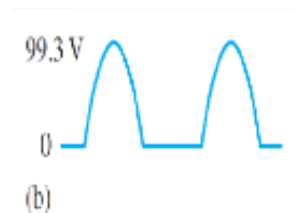
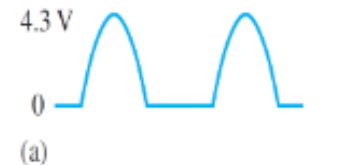


The peak output voltage for circuit (a) is

$$V_{p(out)} = V_{p(in)} - 0.7 \text{ V} = 5 \text{ V} - 0.7 \text{ V} = \mathbf{4.30 \text{ V}}$$

The peak output voltage for circuit (b) is

$$V_{p(out)} = V_{p(in)} - 0.7 \text{ V} = 100 \text{ V} - 0.7 \text{ V} = \mathbf{99.3 \text{ V}}$$



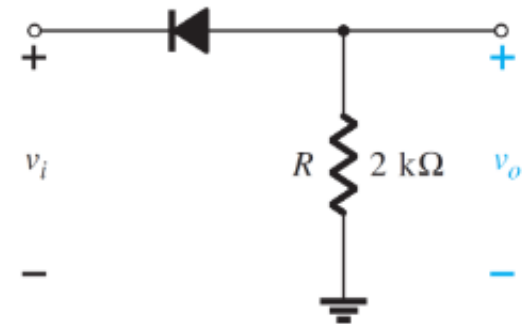
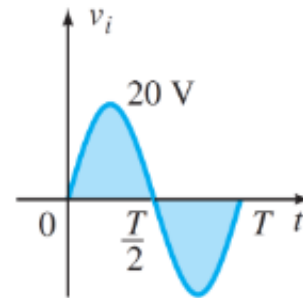
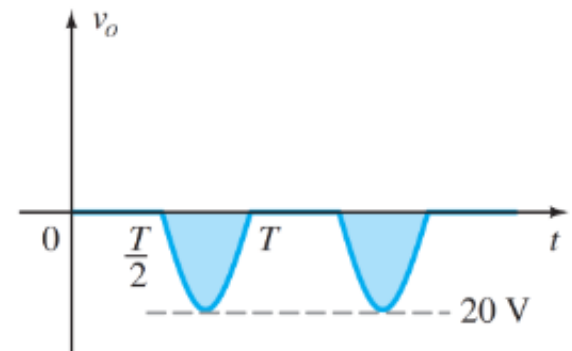
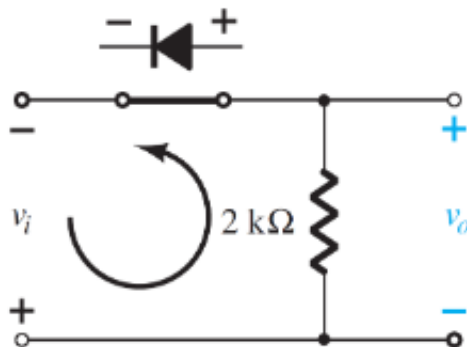
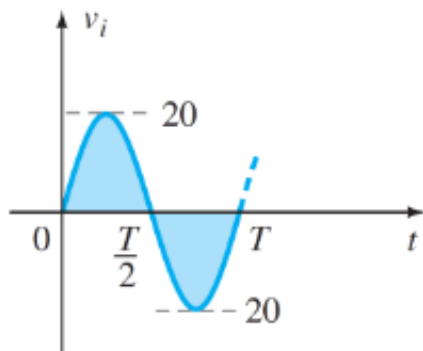
Retificador – Meia Onda

- Example 2**
- Sketch the output v_o and determine the dc level of the output.
 - Repeat part (a) if the ideal diode is replaced by a silicon diode.
 - Repeat parts (a) and (b) if V_m is increased to 200 V.

Solution (a)

- During the positive half cycle the diode is R.B.(O.C.) and $v_o = 0$
- During the negative half cycle the diode is F.B.(S.C.) and $v_o = v_{in}$

$$V_{dc} = -\frac{V_m}{\pi} = -0.318 \times 20 = -6.36V$$

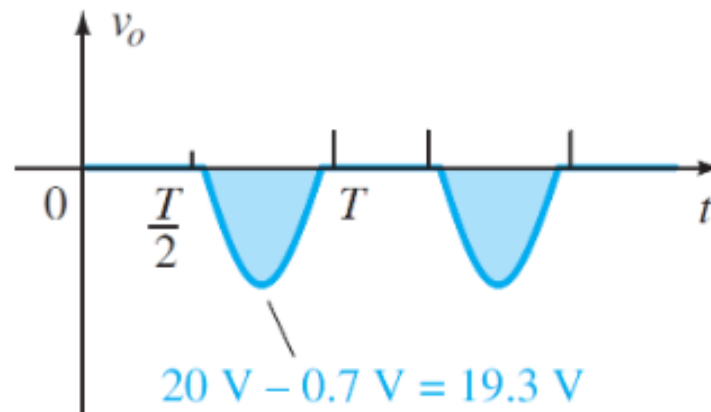


Retificador – Meia Onda

(b) For a silicon diode, the output has the appearance of Fig. below , and

$$V_{dc} \approx -0.318(V_m - 0.7 V) = -0.318(19.3 V) = -6.14 V$$

The resulting drop in dc level is 0.22 V, or about 3.5%.



(c) For $V_m = 200V$

Ideal: $V_{dc} = -\frac{V_m}{\pi} = -0.318 \times 200 = -63.6 V$

Practical: $V_{dc} \approx -0.318(V_m - 0.7 V) = -0.318(199.3 V) = -63.38 V$

which is a difference that can certainly be ignored for most applications

Retificador – Meia Onda

Peak Inverse Voltage (PIV)

- **The peak inverse voltage (PIV)** equals the peak value of the input voltage, and the diode must be capable of withstanding this amount of repetitive reverse voltage.

$$PIV = V_{P(in)}$$

- **The peak inverse voltage (PIV) rating** of the diode is the voltage rating that must not be exceeded in the reverse-bias region or the diode will enter the breakdown region.

$$PIV \text{ rating} \geq V_{P(in)}$$

- A diode should be rated at least 20% higher than the PIV for better performance.

Retificador – Meia Onda

Half-wave rectifier with transformer coupled input voltage.

$$n(\text{Turns ratio}) = \frac{N_{sec}}{N_{pri}}$$

$$V_{sec} = nV_{pri}$$

If $n < 1$ **step down** transformer and $V_{sec} < V_{pri}$

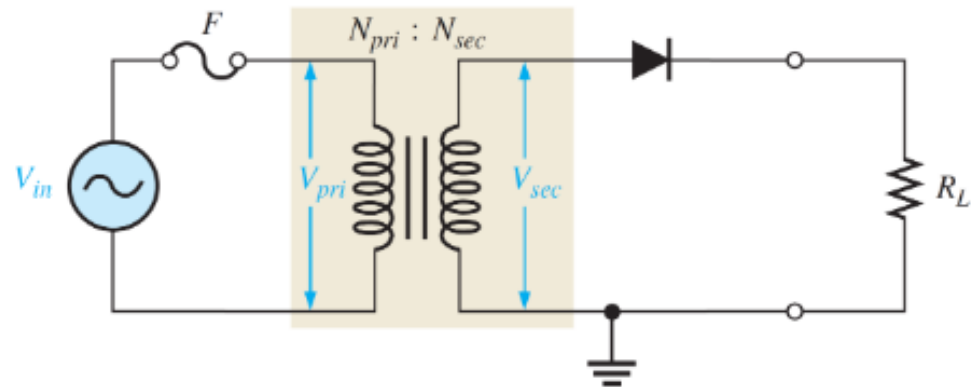
If $n > 1$ **step up** transformer and $V_{sec} > V_{pri}$

The peak output of the rectifier is:

$$V_{p(out)} = V_{p(sec)} - 0.7$$

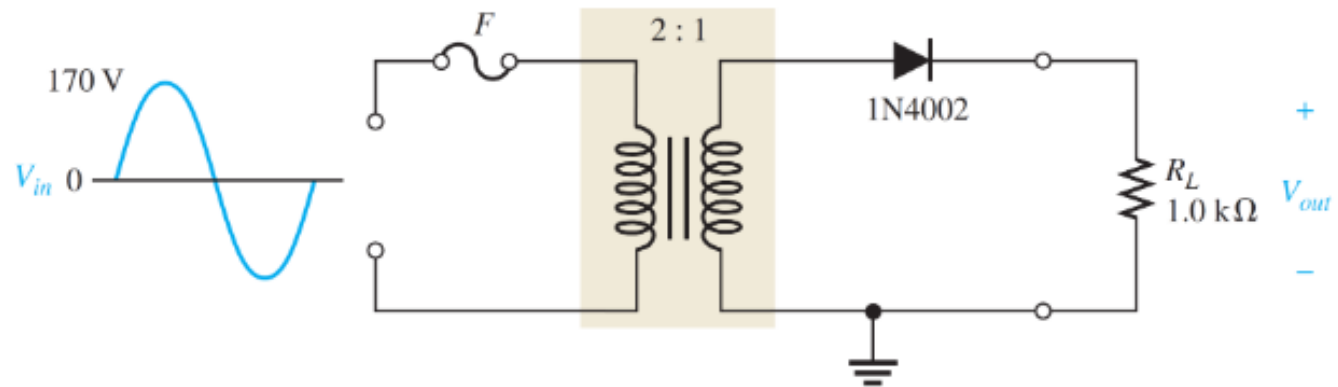
and

$$PIV = V_{p(sec)}$$



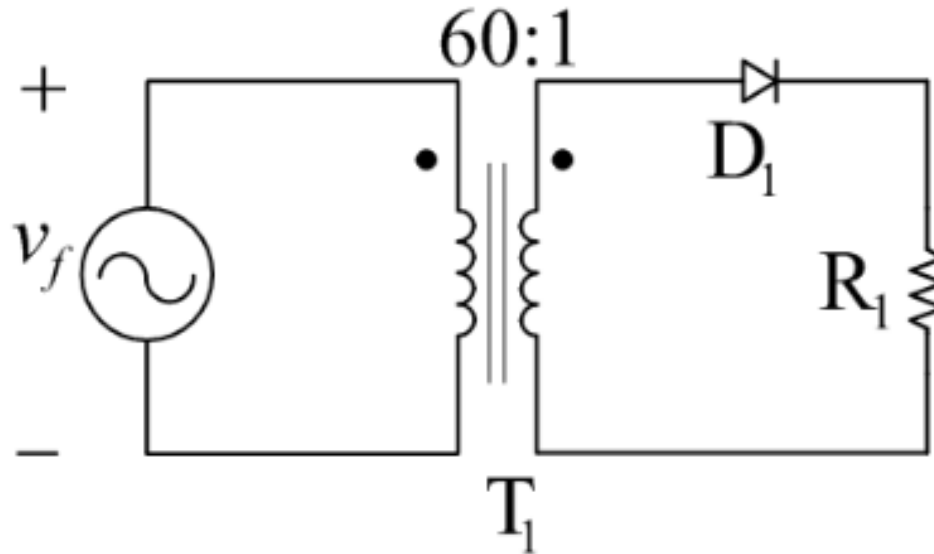
Retificador – Meia Onda

EXAMPLE 3 Determine the peak value of the output voltage.



- $n = \frac{1}{2} = 0.5$
- $V_{p(sec)} = nV_{p(pri)} = 0.5 \times 170 = 85 \text{ V}$
- $V_{p(out)} = V_{p(sec)} - 0.7 = 84.3 \text{ V}$
- $PIV = V_{p(sec)} = 85 \text{ V}$

Meia onda - Exercícios



Considerando os dados ao lado, determine:

- Tensão eficaz no primário de T_1 ;
- Tensão eficaz no secundário de T_1 ;
- Tensão média na saída;
- Tensão de pico na saída;
- Tensão reversa sobre o diodo;
- Corrente média na saída.

$$v_f(t) = 311 \cdot \text{sen}(377 \cdot t) \text{ V};$$

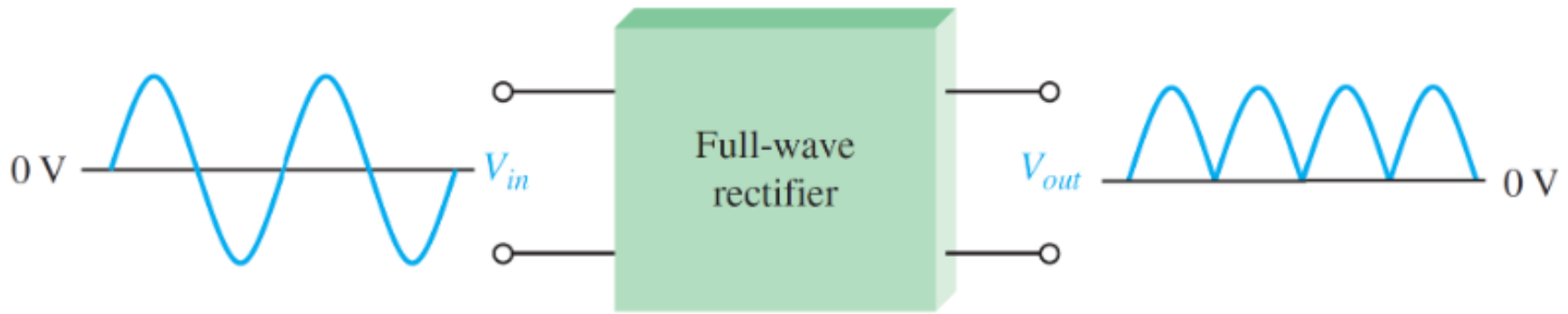
$$R_1 = 5 \, \Omega;$$

$$D_1 = \text{ideal};$$

$$T_1 = 60 : 1.$$

Retificador - Onda Completa

- A **full-wave rectifier** allows unidirectional (one-way) current through the load during the entire cycle of the input, whereas a half-wave rectifier allows current through the load only during one-half of the cycle.
- **FWR** : $f_{out} = 2 f_{input}$ [twice the input frequency]
- $V_{AVG} = \frac{2V_P}{\pi} = 0.637 V_P$ [twice that of HWR]



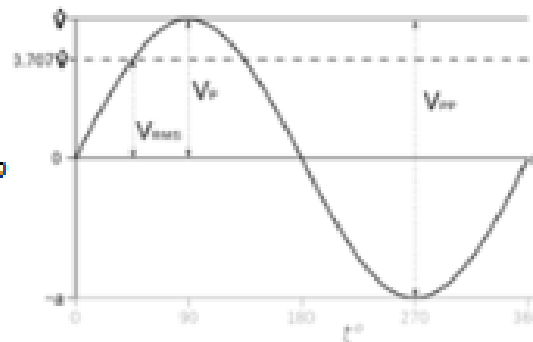
Retificador - Onda Completa

Ondas Senoidais

$V_p = \hat{V}$ = valor de pico

V_{DC} ou V_m = valor DC ou valor médio

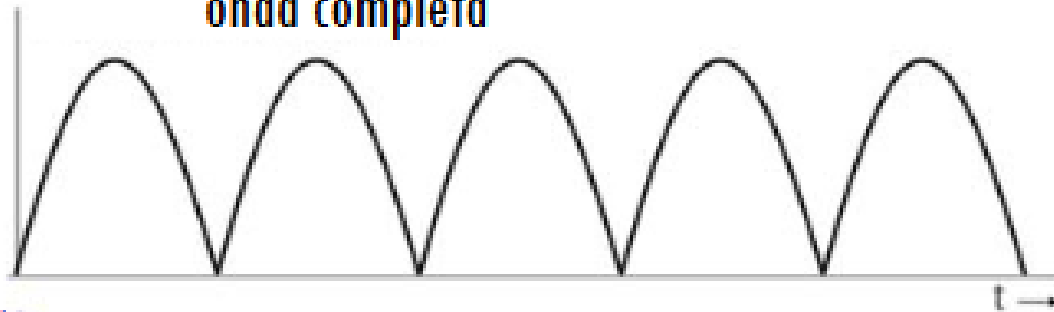
V_{rms} = valor quadrático médio (rms)



$$V_{DC} = \frac{1}{T} \int_0^T V_p \sin(\omega t) dt = 0$$

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^T [V_p \sin(\omega t)]^2 dt} = \frac{V_p}{\sqrt{2}}$$

onda completa

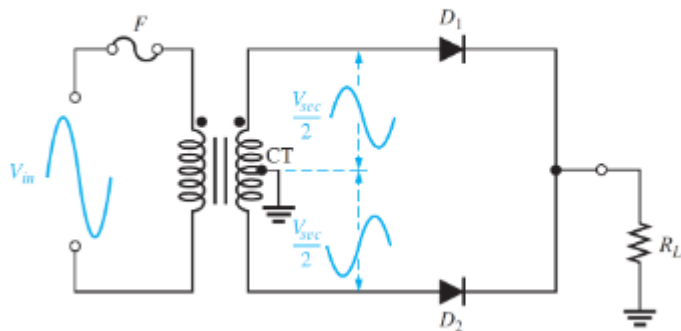


$$V_{DC} = 2 \times \frac{V_p}{\pi}$$

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^T [V_p \sin(\omega t)]^2 dt} = \frac{V_p}{\sqrt{2}}$$

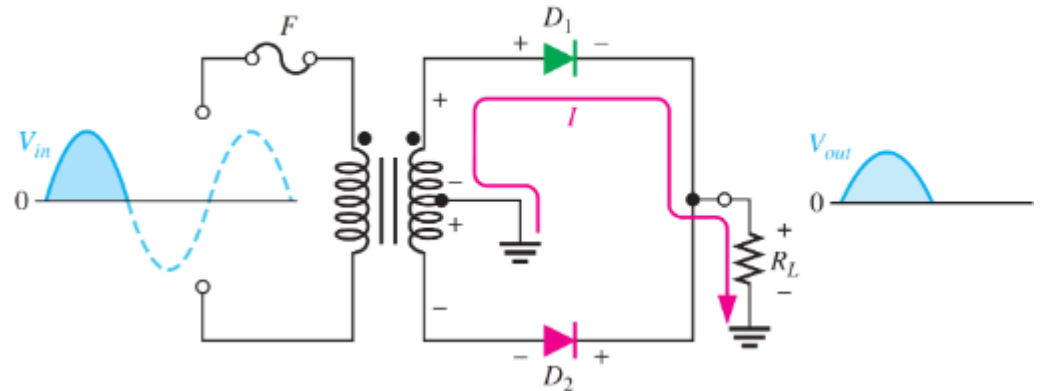
Onda completa Center Tap (CT)

Center-Tapped Full-Wave Rectifier (CT-FWR)

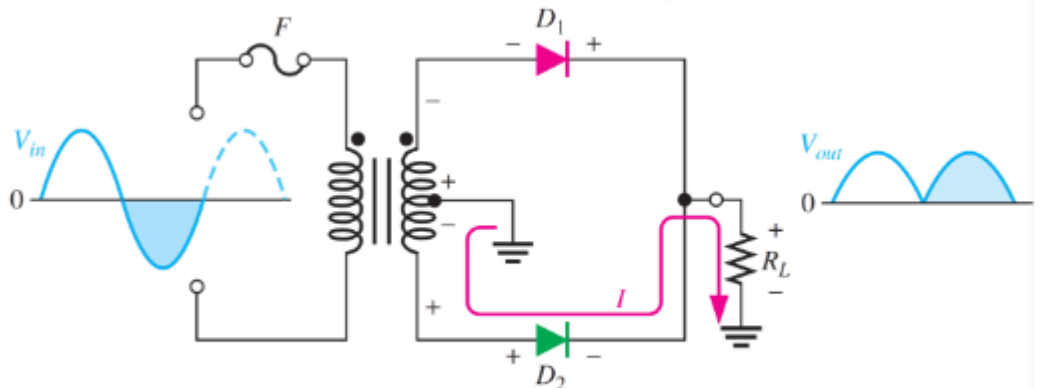


- During the **+ve** half cycle **D1 is ON** and **D2 is OFF**
- During the **-ve** half cycle **D1 is OFF** and **D2 is ON**
- The current passes through R in the same direction in both half cycles

$$V_{out} = \frac{V_{sec}}{2} - 0.7$$



(a) During positive half-cycles, D_1 is forward-biased and D_2 is reverse-biased.



(b) During negative half-cycles, D_2 is forward-biased and D_1 is reverse-biased.

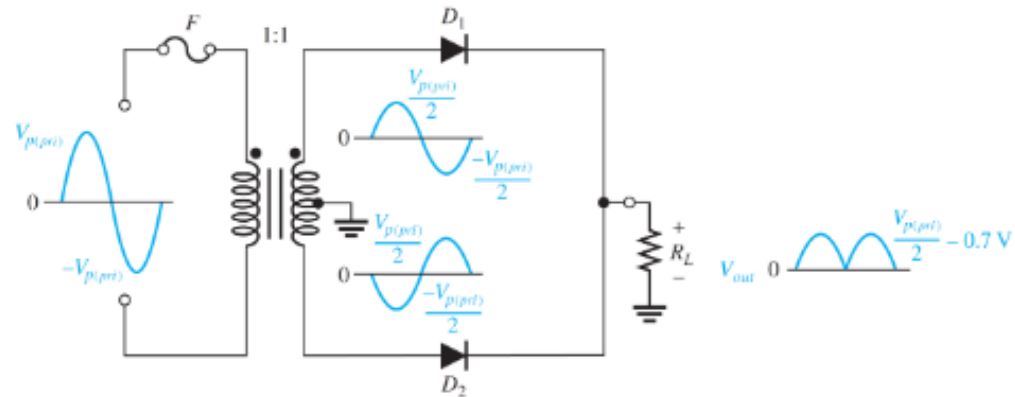
Onda completa Center Tap (CT)

Effect of the Turns Ratio on the Output Voltage

- $n = 1$
- The total secondary voltage is:

$$V_{p(sec)} = V_{p(pri)}$$

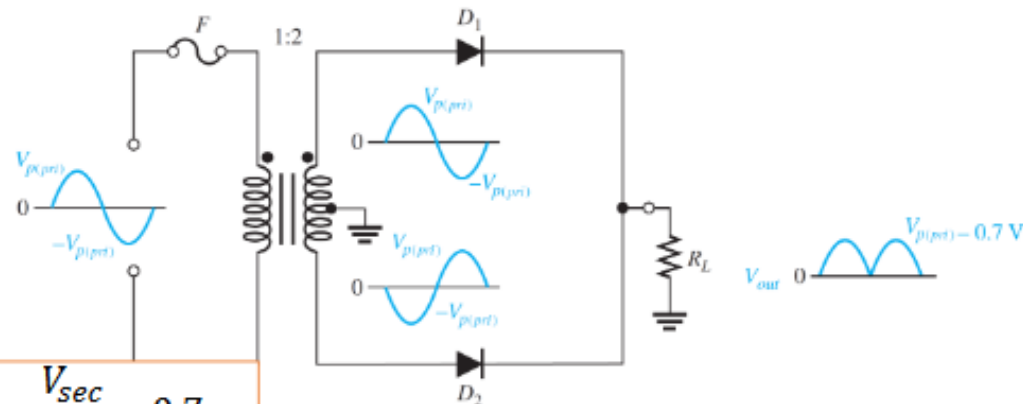
$$\begin{aligned} \bullet V_{p(out)} &= \frac{V_{p(sec)}}{2} - 0.7 \\ &= \frac{V_{p(pri)}}{2} - 0.7 \\ &= \frac{V_{p(input)}}{2} - 0.7 \end{aligned}$$



- $n = 2$
- The total secondary voltage is:

$$V_{p(sec)} = 2 V_{p(pri)}$$

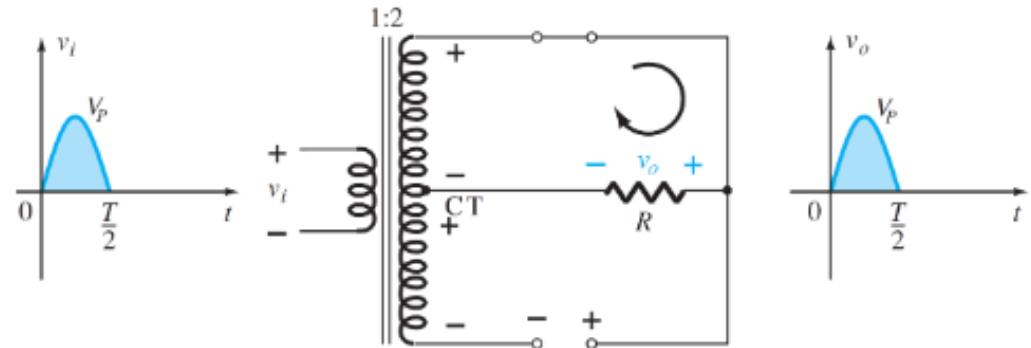
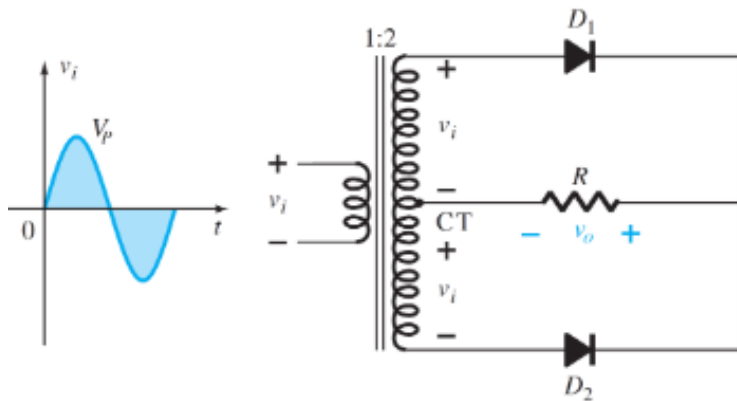
$$\begin{aligned} \bullet V_{p(out)} &= \frac{V_{p(sec)}}{2} - 0.7 \\ &= V_{p(pri)} - 0.7 \\ &= V_{p(input)} - 0.7 \end{aligned}$$



$$V_{out} = \frac{V_{sec}}{2} - 0.7$$

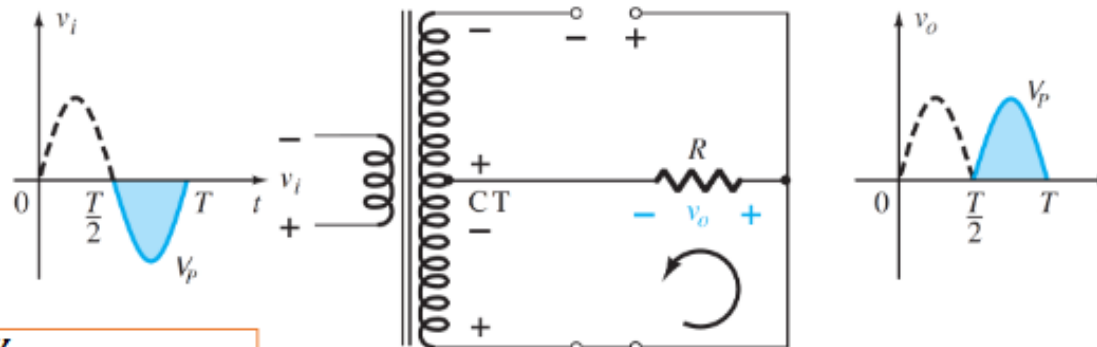
Onda completa Center Tap (CT)

Center-Tapped Full-Wave Rectifier (CT-FWR) [\(Similar configuration\)](#)



Network conditions for the positive region of v_i .

- During the +ve half cycle **D1 is ON** and **D2 is OFF**
- During the -ve half cycle **D1 is OFF** and **D2 is ON**
- The current passes through R in the same direction in both half cycles

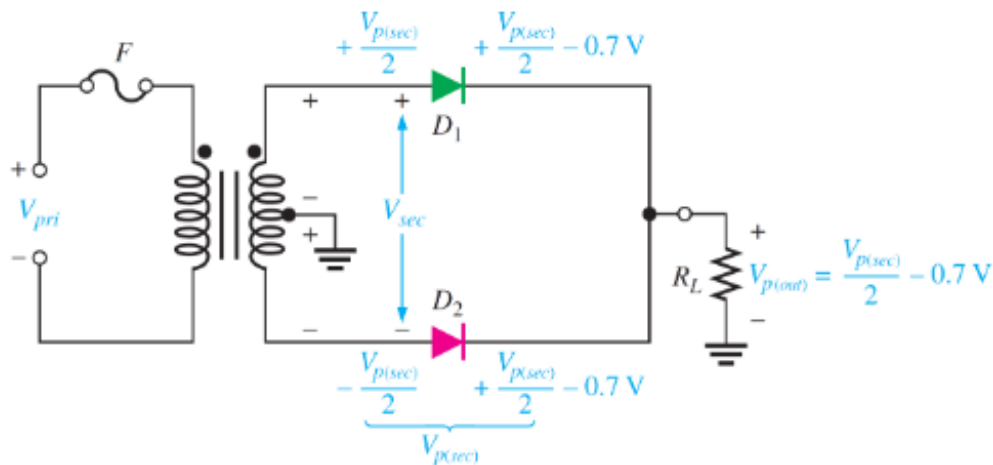


Network conditions for the negative region of v_i .

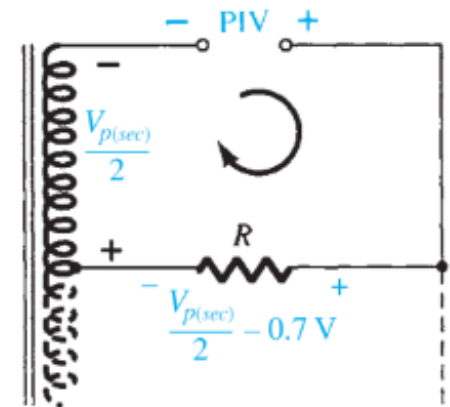
Ideal : $V_{out} = \frac{V_{sec}}{2}$ Practical : $V_{out} = \frac{V_{sec}}{2} - 0.7$

Onda completa Center Tap (CT)

Peak Inverse Voltage



$$\begin{aligned} PIV &= \left(\frac{V_{p(sec)}}{2} - 0.7 \text{ V} \right) - \left(-\frac{V_{p(sec)}}{2} \right) = \frac{V_{p(sec)}}{2} + \frac{V_{p(sec)}}{2} - 0.7 \text{ V} \\ &= V_{p(sec)} - 0.7 \text{ V} \end{aligned}$$



$$\begin{aligned} PIV &= \frac{V_{p(sec)}}{2} + \frac{V_{p(sec)}}{2} - 0.7 \text{ V} \\ &= V_{p(sec)} - 0.7 \text{ V} \end{aligned}$$

$$\boxed{PIV = V_{p(sec)} - 0.7}$$

Onda completa Center Tap (CT)

Example 4

- (a) Show the voltage waveforms across each half of the secondary winding and across R_L
- (b) Find the DC value of the output.
- (c) What minimum PIV rating must the diodes have?

Solution (a) $n = \frac{1}{2} = 0.5$

- The total peak secondary voltage is:

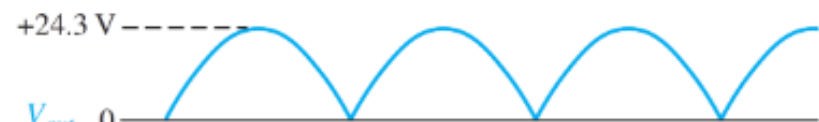
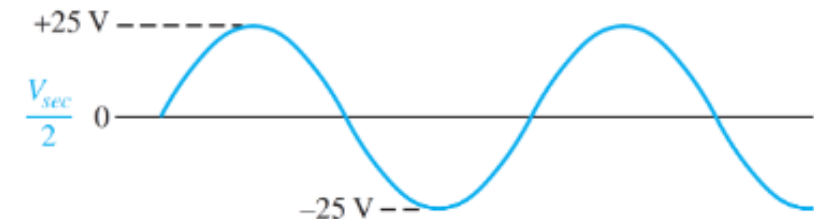
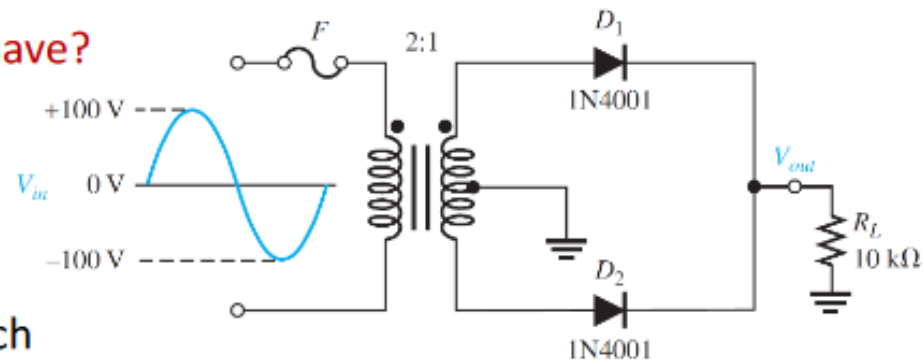
$$V_{p(sec)} = nV_{p(pri)} = 0.5 \times 100 = 50 \text{ V}$$
- Hence There is a $\frac{V_{p(sec)}}{2} = 25 \text{ V}$ peak across each half of the secondary with respect to ground.
- The load voltage has a peak value:

$$V_{p(out)} = \frac{V_{p(sec)}}{2} - 0.7 = 24.3 \text{ V}$$

(b) $V_{AVG} = \frac{2V_{P(out)}}{\pi} = 0.637 V_{P(out)} = 15.48 \text{ V}$

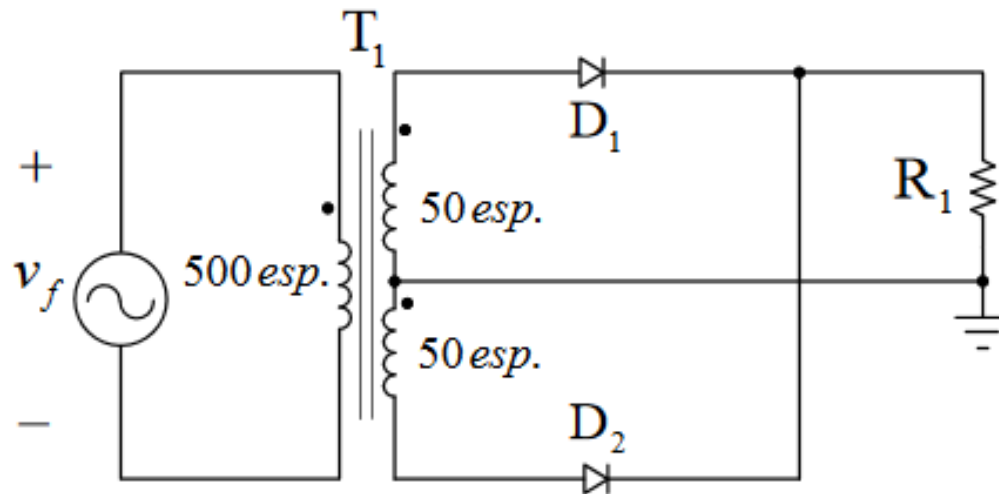
- (c) Each diode must have a minimum PIV rating of

$$PIV = V_{p(sec)} - 0.7 = 49.3 \text{ V}$$



Onda completa CT - Exercício

Considerando o circuito abaixo:



Considerando os dados ao lado, determine:

- Tensão eficaz no primário de T_1 ;
- Tensão eficaz no secundário de T_1 ;
- Tensão média na saída;
- Tensão de pico na saída;
- Tensão reversa sobre os diodos;
- Corrente média na saída.

$$v_f(t) = 311 \cdot \sin(377 \cdot t) V;$$

$$R_1 = 5 \Omega;$$

$$D_{1_2} = \textit{ideais};$$

$$T_1 = \begin{cases} 10:1 \\ 10:1 \end{cases}$$

Tarefas

- Simule os circuitos apresentados nos slides:
- 20, 25, 26, 32 e 33

Disciplina: Eletrônica Básica

Aula_4 - Diodos

Ambiente Blackboard